VR Project 2

Multimodal Visual Question Answering with Amazon Berkeley Objects Dataset

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As part of this project, we created a multiple-choice Visual Question Answering (VQA) dataset using the Amazon Berkeley Objects (ABO) dataset, evaluated multiple baseline models, applied fine-tuning through Low-Rank Adaptation (LoRA), and finally measured performance using standard evaluation metrics.

The ABO dataset contains 147,702 product listings with multilingual metadata and 398,212 unique catalog images. We used the small variant (3GB) - abo-images-small.tar - from the downloads section to avoid size issues; it includes metadata in CSV format and 256x256 resolution images.

More metadata about each of the images were also provided in a separate download - abo-listings.tar which contained several JSONL files that had more expressive metadata about each image such as bullet points, its dimensions, language tags etc.

Dataset Structure :

The images folder had subfolders named from 00 to ff. Within each folder there were multiple JPEG images, each with a unique filename and a metadata.csv outside all these folders that linked the image paths to a particular image id.

Similarly in the listings folder there were listings JSONL files where each row of any of these files were a valid JSON object with a main_image_id and other_image_ids (array) fields.

In this way, one can connect an image by its filename to its record in metadata.csv to retrieve the image_id and link that to the corresponding json objects in any of the JSONL files.

Dataset Curation :

Thus, we created our data curation scripts that would take a certain subfolder of the images folder, create a static lookup (dictionary with image filename as key and JSON object as value) for each of its images, connecting each to its JSON metadata.

Then we used **Google Gemini 2.0 Flash** (a large multimodal foundation model) to generate multiple-choice Visual Question Answering (VQA) questions from the images and metadata by passing suitable prompts.

Input to Gemini included

- Raw images from the ABO dataset.
- Associated JSON metadata (attributes, categories, descriptions).
- Prompt passed

We used 2 prompts for this task for each image. The prompts were :

- You are a VQA system. Given an image and metadata, generate an EASY question grounded in visible content. Return a JSON with: question and answer (answer should be a single word only)

- You are a VQA system. Given an image and metadata, generate a HARD question focusing on a specific part or feature of the image. Return a JSON with: question and answer (answer should be a single word only).

By generating both easy and hard questions for each image, we simulate a curriculum learning approach where easy questions test basic object recognition and scene understanding & hard questions probe for fine-grained understanding, specific regions, or nuanced attributes.

Easy prompt captures general content, common objects, scene-level features like colours and shapes and identity of the object.

The hard prompt forces the model to focus on specific parts, intricate details, or rare attributes and even reason, like identifying text on the object, or identifying which object has the smallest opening etc.

With a daily quota of 1000 requests only, we spread this task over a few days incrementally collecting data till we amassed ~8.5K questions to fine tune the baseline models (with LoRA) and ~3K questions to test their baseline performance and again after compressing them & fine tuning them.

Baseline Evaluation:

We then decided to use 3 off the shelf VQA models to create our initial benchmarks. The ones we ended up selecting were ViLT, BLIP and BLIP-2. We also considered other alternatives such as CLIP, ViLBERT and LLaVA.

However, CLIP works best as a feature extractor and is not designed for VQA, especially as it was single word answers and not MCQ.

ViLBERT was disregarded mainly because of its huge model size and number of parameters as well as it learning the text and image encodings separately and thus it is both slower and worse than newer unified models like BLIP or LLaVA.

LLaVA while being very powerful was also disregarded because of its large size, demands lot of computational resources and the fact that its better suited to free form text generation than precise 1 word answers.

ViLT, BLIP and BLIP-2 struck a good balance between being lighter models that were also faster, more integrated learning text and image representations and suited for VQA answer generation.

We tested them all, without fine tuning, on our test VQA subset and below were the results :

Baseline Evaluation Metrics :

	Accuracy	F1-Score	BERT-Score	BART-Score	Rouge-L
ViLT	0.0001	0.0001	0.9578	-5.6399	0.3257
BLIP	0.4453	0.1143	0.9702	-4.6868	0.4474
BLIP-2	0.1701	0.0376	0.8994	-6.5398	0.1275

Fine Tuning and Model Compression:

We decided to use Quantization to compress the models and LoRA to fine tune the models on the ABO VQA dataset. These 2 techniques are well suited to work with each other, often called QLoRA because of it.

Quantization compresses models by reducing weights and activations from 32-bit floats to lower-bit formats. In our scripts we reduced them to 4-bit formats.

Quantization results :

ViLT : Model size reduced from 470MB to 220MB
BLIP : Salesforce/blip-vqa-base variant reduced size from 1.5GB to ~400 MB. Effective parameter count reduced from ~380 million to ~230 million
BLIP-2 : Salesforce/blip2-flan-t5-xl variant reduced size from 15GB to ~2GB. Effective parameter count reduced from 3 billion parameters to around 2 billion parameters.

Then after we had quantized the base models, we applied LoRA with similar configs for all of them to fine tune them. Rank chosen was 8, Alpha was 32, with dropout of 0.05 and target modules being attention layers.

In BLIP, the vision encoder was not touched, but all the self and cross attention layers of both the text encoder and decoder were targeted. In BLIP-2, the last 6 layers of the Q-Former and the last 6 blocks of the T5 language model's encoder and decoders were targeted with LoRA. Similarly in VilT the self attention layers were targeted.

A lot of challenges were encountered during the fine-tuning. For BLIP and BLIP-2, direct use of Hugging Face Trainer module was proving to be impossible because it would always internally pass an extra parameter inputs_embeds to the base

BlipForQuestionAnswering module’s forward method. As such we manually patched the functionality by describing a custom subclass to BlipForQuestionAnswering that overrode the forward method, popped the extra argument, and passed the correct argos to the superclass forward method. Similarly, for BLIP, we also had to override addition of positional embeddings because of errors in the inbuilt function which came down to how Python (and PyTorch) handle in-place operations and tensor gradients. Then during both ViLT and BLIP-2 finetuning, we experienced more training difficulties with training loss becoming NaN with gradient explosion that we controlled with gradient clipping for BLIP-2 but VilT finetuning was unsuccessful for us.

Fine Tuned Evaluation Metrics :

	Accuracy	F1-Score	BERT-Score	BART-Score	Rouge-L
ViLT					
BLIP	0.4385	0.1027	0.9693	-4.6134	0.4414
BLIP-2	0.1526	0.0021	0.8923	-5.5145	0.1166

Extra Evaluation Metrics chosen :

Other than direct token comparison metrics like Accuracy and F1-Score, we decided to use NLP based semantic similarity measures like BERT-Score (suitable for VQA as it captures semantic similarity between predicted and ground truth answers), BART-Score (suitable for VQA than lexical metrics as it uses a pretrained language model to assess answer relevance and fluency) and Rouge-L (less suitable for VQA as it focuses on sequence overlap, not factual correctness and struggles with evaluating short, precise answers) to compare our generated answers to ground truth answers.